AYMANE AIT DADS

Data Science Intern | Machine Learning · Python · Analytical Problem-Solving

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PROFESSIONAL SUMMARY

Data Science engineering student at EURECOM who reduced manual network analysis time by ~60% at Orange Maroc by applying machine learning models to 100K+ fiber-optic records and delivering actionable insights to a network optimization team. Built automated clustering pipelines mapping customer performance profiles to 5 commercial service tiers, translating complex model outputs into stakeholder-ready Power BI dashboards and maintenance recommendations. Brings a solid understanding of machine learning techniques and algorithms, expert Python programming, and hands-on experience designing and optimizing end-to-end data science solutions - including a top-17 finish out of 3,000+ teams in an ongoing Kaggle competition ($106K+ prize pool) demonstrating exceptional technical depth. Aims to collaborate with the data science team at Innovative Technology to design and implement solutions that turn raw data into measurable business impact.

CORE COMPETENCIES

Machine Learning Techniques | Python Programming | Data Science | Analytical Problem-Solving | Algorithm Design | Model Optimization | Collaborative Teamwork | Communication of Insights

WORK EXPERIENCE

Orange Maroc — *Jul 2024 - Aug 2024*

Data Science Intern - Casablanca, Morocco

- Designed and implemented Random Forest and K-Means machine learning models on **100K+ fiber-optic network records** (upload speed, latency, jitter) to classify network quality across thousands of nodes.
- Developed an automated clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for operational decisions.
- Reduced manual analysis time by **~60%** through automated feature engineering and model evaluation pipelines, freeing team capacity for higher-value analytical problem-solving tasks.
- Delivered stakeholder presentations translating machine learning model outputs into **Power BI** dashboards and concrete maintenance recommendations, bridging technical findings with business objectives.

PROJECTS

Aerial Cactus Detection - Endangered Species Classification — *EURECOM Course Project · 2025 · 99.8% Accuracy*

- Benchmarked CNN, DenseNet121, and hybrid CNN-Transformer architectures on **17,500+ drone images** to classify endangered cactus species, achieving 99.8% accuracy, F1 = 0.999, and ROC AUC = 1.0.
- Designed and optimized an end-to-end image classification pipeline using ensemble methods and rigorous model evaluation, applying analytical problem-solving to select the highest-performing algorithm per metric.

PyTorch · DenseNet121 · Scikit-learn · CNN

NVIDIA Nemotron Model Reasoning Challenge — *Kaggle Competition · In Progress (Jun 2026) · $106K+ Prize Pool*

- Currently ranked **17th / 3,000+ teams** (score 0.86/1.0, $106K+ prize pool) in the ongoing NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter** model with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context).
- Ran **controlled ablation experiments** across learning-rate schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers for verified chain-of-thought data.

PyTorch · Unsloth · PEFT · Hugging Face Transformers · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM — *Sep 2023 - Present*

Relevant coursework: Machine Learning, Deep Learning, Statistics, Computer Vision, NLP, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya — *2021 - 2023*

SKILLS

Data Science & Analytics: Python | SQL | Pandas | NumPy | Power BI | Scikit-learn | Feature Engineering

Machine Learning & Deep Learning: PyTorch | Random Forest | K-Means Clustering | Ensemble Methods | TensorFlow | Hugging Face Transformers

Tools & Visualization: Matplotlib | Seaborn | Git | Linux | Jupyter Notebooks | Bash